

Gravitational fields

A Level Physics

Gravitational fields

Definition

A **gravitational field** 重力场 is a region where a **mass** 质量 feels a **force** 力 from other masses. The **gravitational field strength** 重力场强度 g at a point is the **gravitational force per unit mass** on a small **test mass** 检验质量 placed there:

$$g = \frac{F}{m}.$$

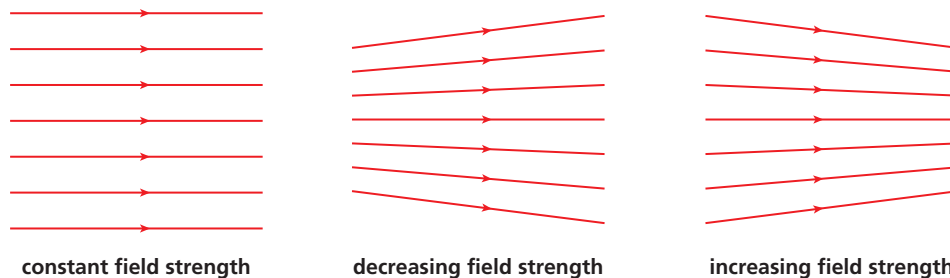
Unit: N kg^{-1} (the same as m s^{-2} —the acceleration of free fall in the field). g is a **vector** 矢量, pointing the way the force acts —towards the source mass.

Field lines

A gravitational field is drawn with **field lines** 场线 that point the way the force acts on a test mass:

- around a **point mass** 质点 or a uniform sphere (treated as a point mass from outside), the field lines are **radial** 径向, pointing **inwards**.
- near the Earth's surface over a small area, the field lines are nearly **parallel and equally spaced**, pointing straight down —a **uniform field** 匀强场.

Closer lines mean a stronger field.



Field-line spacing shows the field strength —closer lines mean a stronger field

Newton's law of gravitation

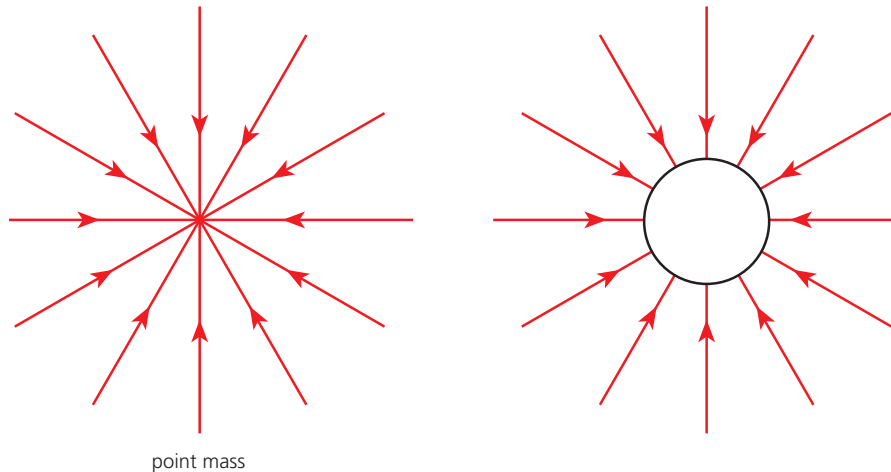
For two point masses m_1, m_2 a distance r apart, the force on each is

$$F = \frac{Gm_1m_2}{r^2},$$

pulling them together along the line joining them. This is **Newton's law of gravitation** 万有引力定律. The constant $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ is the **universal gravitational constant** 万有引力常量.

Spheres treated as point masses

For a uniform sphere (such as a planet or star), the field at any point **outside** is the same as that of a point mass equal to the total mass at the centre. So from above the surface, you can treat the Earth as a point mass at its centre. (Points inside a sphere are different, and are not in the syllabus.)



Outside a uniform sphere the field is radial, exactly like a point mass at the centre

Field strength from a point mass

Put the gravitational force on a test mass m at distance r from a point mass M into $g = F/m$:

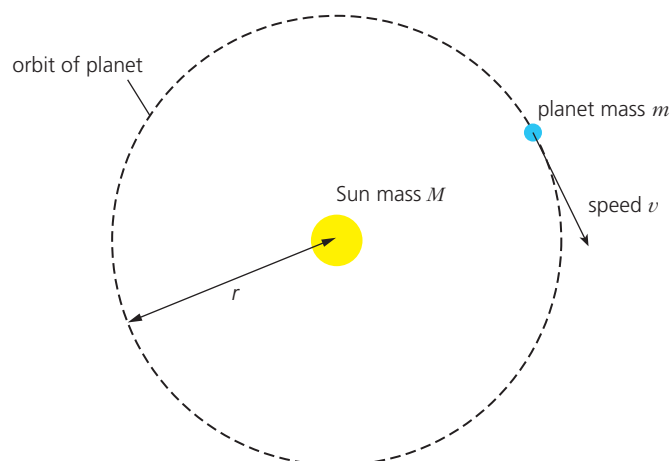
$$F = \frac{GMm}{r^2}, \quad g = \frac{GM}{r^2}.$$

So g falls off as $1/r^2$ as you move away from the source.

Why g is nearly constant near the Earth's surface

The Earth's radius is $R \approx 6.4 \times 10^6$ m. Rising to height h changes the distance from the centre from R to $R + h$. For $h \ll R$ (any building or mountain), $(R + h)/R \approx 1$, so g barely changes—going from 5 m to 10 m high changes r by about one part in a million. In the laboratory, g is effectively constant.

Orbital motion in a gravitational field



Gravity provides the centripetal force that keeps a planet in a circular orbit

For a **satellite** 卫星 of mass m in a circular **orbit** 轨道 of radius r around a body of mass M , gravity provides the **centripetal force** 向心力:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

Cancel m (the orbital speed does not depend on the satellite's mass):

$$v = \sqrt{\frac{GM}{r}}.$$

The **period** 周期 follows from $T = 2\pi r/v$:

$$T = 2\pi\sqrt{\frac{r^3}{GM}}, \quad \text{so} \quad T^2 = \frac{4\pi^2}{GM} \cdot r^3.$$

This is **Kepler's third law** 开普勒第三定律 for circular orbits: $T^2 \propto r^3$. A plot of T^2 against r^3 is a straight line through the origin with gradient $4\pi^2/(GM)$, so orbital data gives the central mass.

Geostationary orbit

A **geostationary** 地球同步 satellite:

- stays directly above the same point on the Earth (so a fixed dish always points at it),
- has a period of **24 hours** (the same as the Earth's rotation),
- orbits **west to east** (the same way the Earth turns),
- must be directly above the **equator** 赤道.

It must have the same **angular speed** 角速度 as the Earth, in the same direction, in the equatorial plane (or it would drift north-south during the day). From $T = 24$ h and $T^2 = 4\pi^2 r^3/(GM)$, the radius is $r \approx 4.2 \times 10^7$ m (about 3.6×10^7 m above the surface).

Gravitational potential

Gravitational potential 引力势 ϕ at a point is the **work done per unit mass** in bringing a small test mass from **infinity** 无穷远 to that point:

$$\phi = \frac{W}{m}.$$

Unit: J kg^{-1} .

The potential is taken as **zero at infinity**. As the test mass falls in towards the source, gravity does the work for you, so ϕ is **negative** everywhere except at infinity. For a point mass M at distance r :

$$\phi = -\frac{GM}{r}.$$

ϕ is a **scalar** 标量. For several masses, add the potentials.

Gravitational potential energy of two point masses

If a test mass m sits where the potential is ϕ , the **gravitational potential energy** 重力势能 of the pair is

$$E_P = m\phi = -\frac{GMm}{r}.$$

Like the potential, E_P is **negative** and reaches zero only at infinite separation. Closer masses have more negative potential energy (more tightly bound).

Link with $\Delta E_P = mg\Delta h$

For small height changes near the surface, r barely changes, so $\Delta E_P \approx mg\Delta h$. For large changes (a satellite moving to a higher orbit) use $-GMm/r$ at each radius and take the difference:

$$\Delta E_P = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad (r_2 > r_1),$$

which is positive (energy must be supplied to raise the satellite).

Escape velocity (from conservation of energy)

To escape from radius r to infinity, an object's **kinetic energy** 动能 must equal the size of its gravitational potential energy:

$$\frac{1}{2}mv_{\text{esc}}^2 = \frac{GMm}{r}, \quad v_{\text{esc}} = \sqrt{\frac{2GM}{r}}.$$

At the Earth's surface, the **escape velocity** 逃逸速度 is $\approx 11 \text{ km s}^{-1}$. It does not depend on the object's mass.